

Lecture 3

Understanding op-amp specifications

Prof Peter YK Cheung
Imperial College London

URL: <https://elec50001-cas.github.io/course-website/>
E-mail: p.cheung@imperial.ac.uk

What you should know already?

Imperial College
London

EE1 Analogue Electronics

C-E Amplifier Revisited Quiescent Analysis

- KVL on input side

$$V_{BIAS} = I_B R_B + V_{BE}$$

and $V_{BE} \approx 0.7 \text{ V}$, $I_E =$

$$\Rightarrow I_E \approx \frac{V_{BIAS} - V_{BE}}{R_E}$$

- As before:

$$V_E = I_E R_E$$

Imperial College
London

EE1 Analogue Electronics

BJT Current Mirror

INPUT SIDE



- For transistors identical apart from

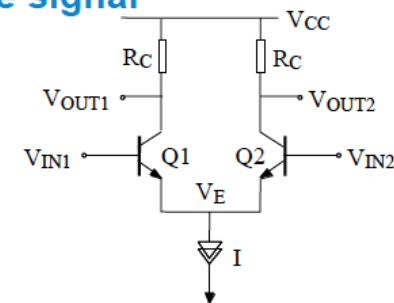
$$I \approx I_{REF}$$

where we have used $I_S \propto A$, and

Imperial College
London

EE1 Analogue Electronics

The Differential Pair – large signal



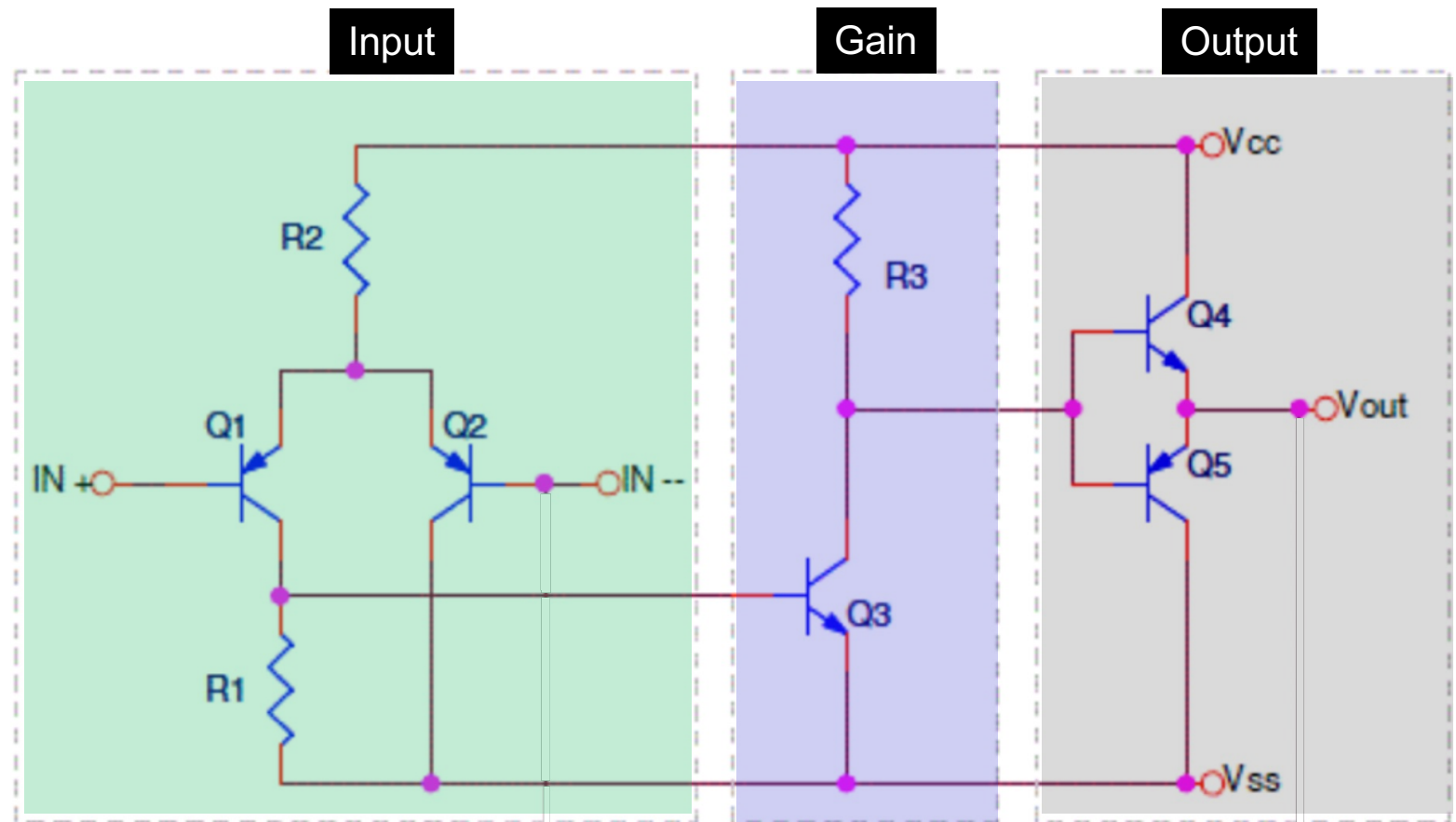
- If transistors are matched, then:

$$I_{C1} = \frac{I}{1 + \exp(-V_D/V_T)} \quad I_{C2} = \frac{I}{1 + \exp(V_D/V_T)}$$

$V_D = (V_{IN1} - V_{IN2})$ is the DIFFERENTIAL INPUT VOLTAGE

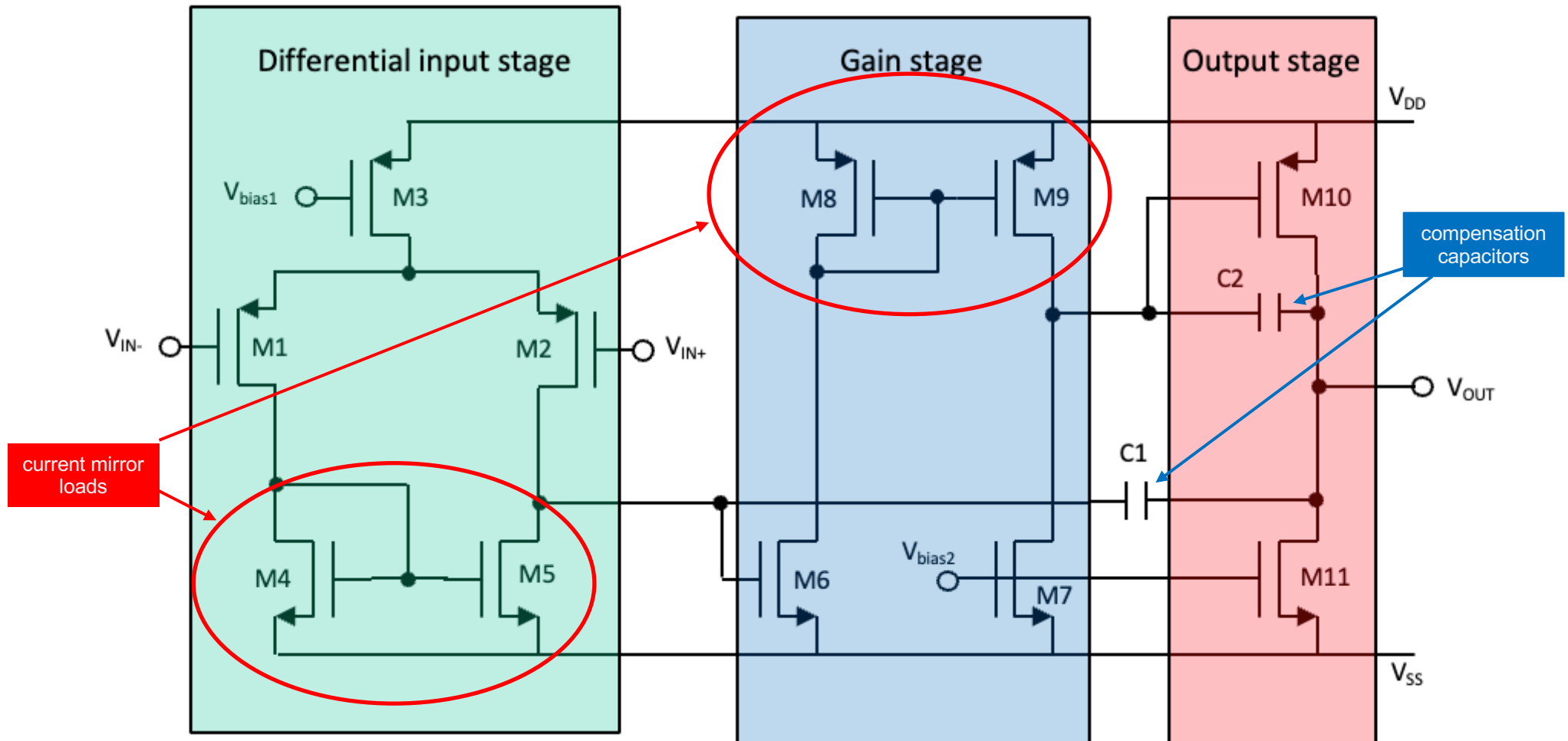
Inside a typical BJT op-amp

- ❖ Three stages architecture:
 1. **Differential input stage** – long-tail pair (Yr 1 Circuits part 2, adc_9, slides 8-14)
 2. **Voltage gain stage** – common emitter amp (adc_6, slides 3-7)
 3. **Output drive stage** – push-pull circuit



Inside a typical MOSFET op-amp

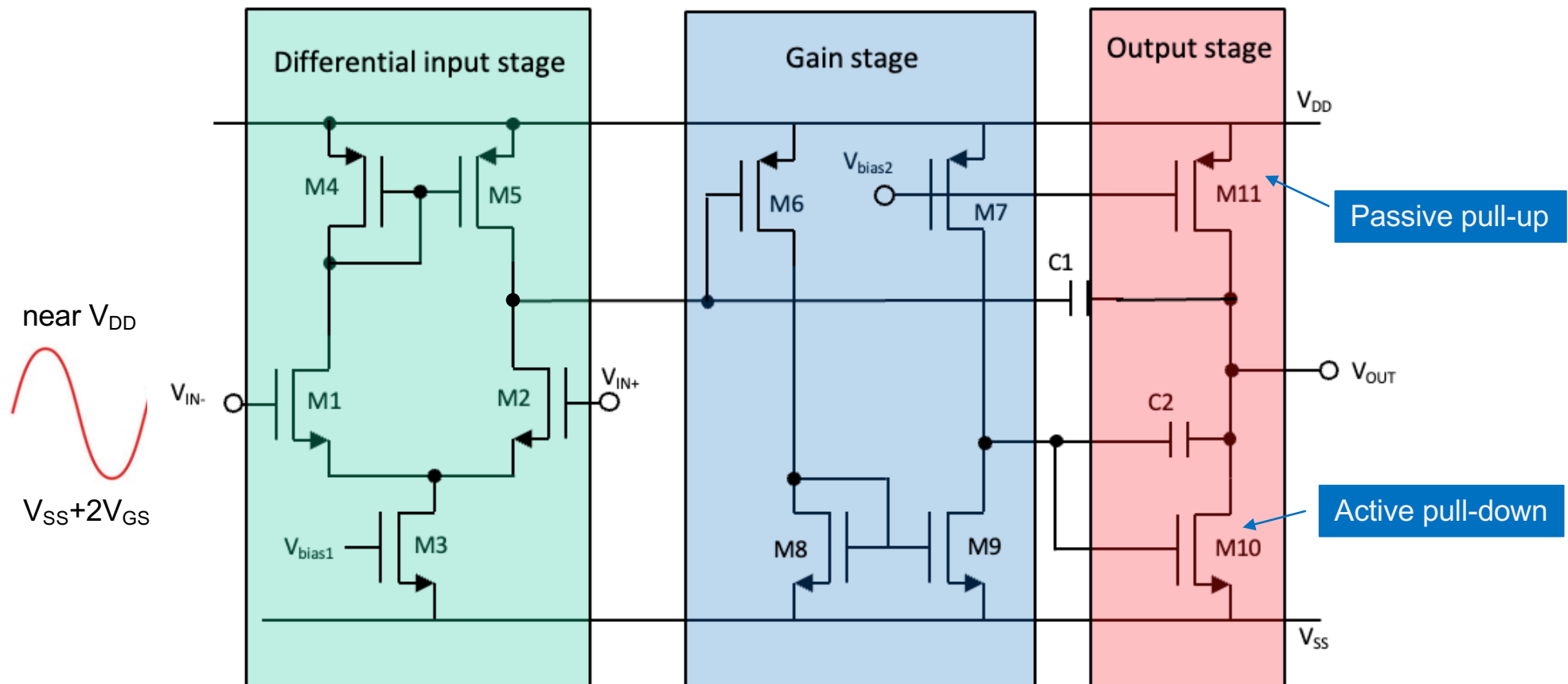
- ❖ Early MOSFET op-amp follows similar architecture to the BJT version.



n-type input MOSFETs but not rail-to-rail

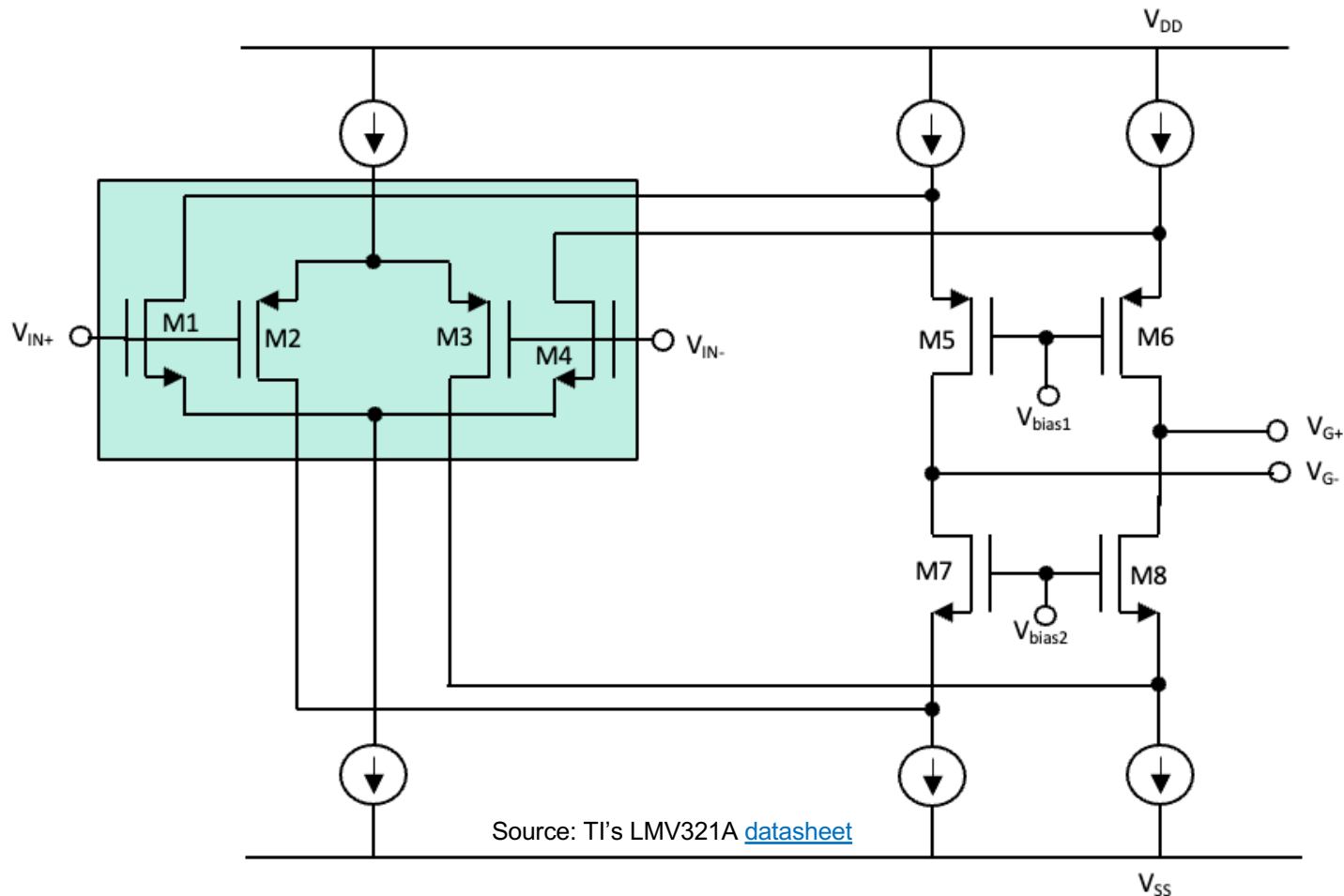
- ❖ Similar op-amp is obtained by:
 - Flip circuit up-side-down.
 - Replace all n-type with p-type and vice versa.

- Not rail-to-rail input or output.
- Asymmetrical output drive.
- Bad for low-voltage, single supply.



Complementary differential input

- ❖ Use complementary differential input to solve input rail-to-rail issue.



Common-Mode Input Range	V_{CMR}	$V_{SS} - 0.3$	—	$V_{DD} + 0.3$	V
-------------------------	-----------	----------------	---	----------------	---

Source: MCP6001 [datasheet](#)

Op-amp Input Specifications

- ❖ These are the input specifications for the MCP6001 op-amp.

Input Offset						
Input Offset Voltage	V_{OS}	-4.5	—	+4.5	mV	$V_{CM} = V_{SS}$ (Note 1)
Input Offset Drift with Temperature	$\Delta V_{OS}/\Delta T_A$	—	± 2.0	—	$\mu V/^{\circ}C$	$T_A = -40^{\circ}C$ to $+125^{\circ}C$, $V_{CM} = V_{SS}$
Power Supply Rejection Ratio	PSRR	—	86	—	dB	$V_{CM} = V_{SS}$

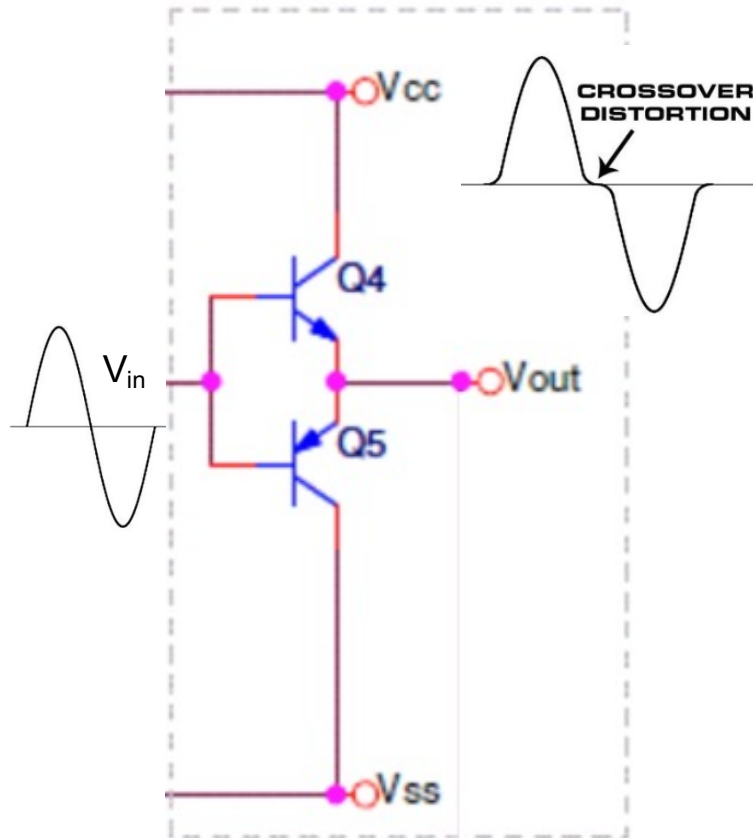
Input Bias Current and Impedance						
Input Bias Current:	I_B	—	± 1.0	—	pA	$T_A = +85^{\circ}C$
Industrial Temperature	I_B	—	19	—	pA	
Input Offset Current	I_{OS}	—	± 1.0	—	pA	
Common-Mode Input Impedance	Z_{CM}	—	$10^{13} 6$	—	ΩpF	
Differential Input Impedance	Z_{DIFF}	—	$10^{13} 3$	—	ΩpF	

Common-Mode						
Common-Mode Input Range	V_{CMR}	$V_{SS} - 0.3$	—	$V_{DD} + 0.3$	V	
Common-Mode Rejection Ratio	CMRR	60	76	—	dB	$V_{CM} = -0.3V$ to $5.3V$, $V_{DD} = 5V$

 Highlighted entries are implemented in the MCP6001 LTSpice model.

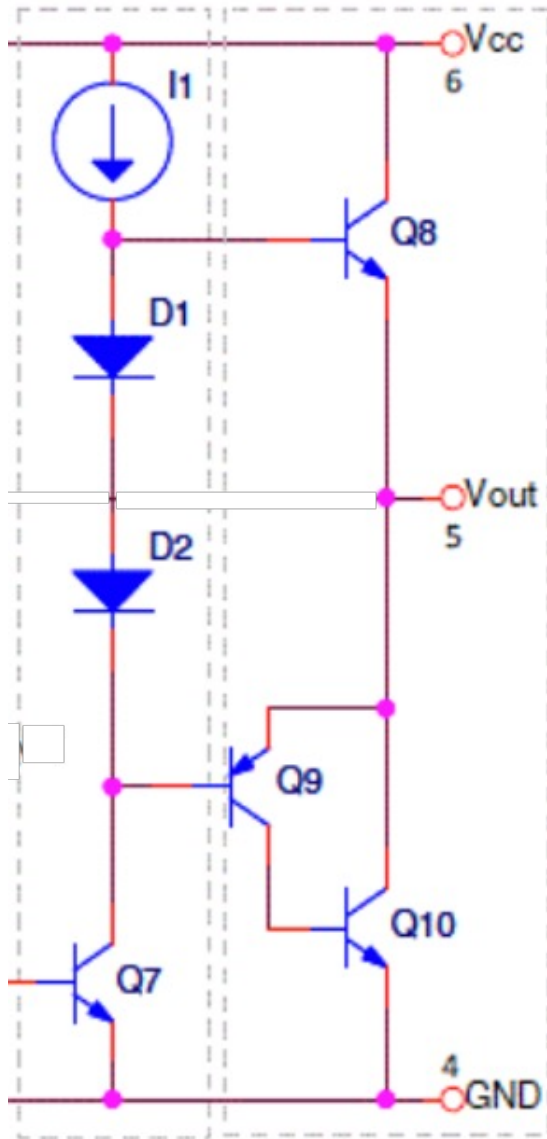
Source: MCP6001 [datasheet](#)

Output Stage – Class B



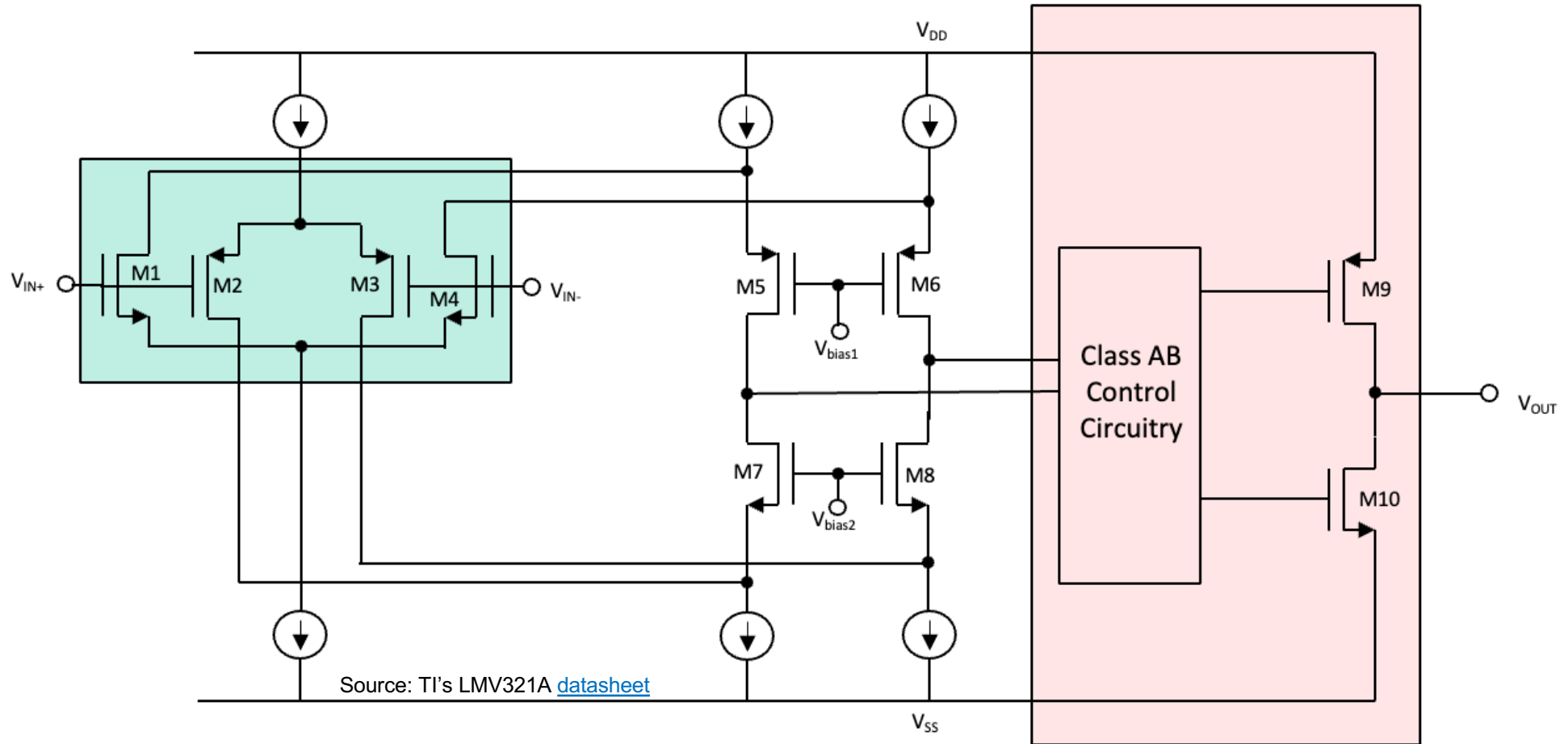
- ❖ Yr 1st ADC part 2 Lecture 6, S3-5
- ❖ Q4 is emitter follower (Common-Emitter) for sourcing current to V_{out} (PUSH)
- ❖ Q5 is another emitter follower for sinking current from V_{out} (PULL)
- ❖ This is known as a PUSH-PULL or class B amplifier circuit
- ❖ $\delta V_{out} \approx \delta V_{in}$, i.e. its gain is 1
- ❖ Each transistor only operate for half cycle or 180° of a sinewave signal
- ❖ Further, Q4 and Q5 requires $V_{BE} > 0.7V$ to start conducting, therefore this amplifier has distortion.

Bipolar output Stage – Class AB output



- ❖ Q_8 push emitter follower sourcing current
- ❖ Q_9 , Q_{10} pull emitter follower sinking current
- ❖ D_1 , D_2 forward bias due to I_1
- ❖ Keep Q_8 and Q_9 in linear region – reduce distortion
- ❖ Q_9 PNP has poor current gain
- ❖ Combine with Q_{10} NPN to boost current gain
- ❖ Q_9 , Q_{10} - called a “**Sziklai**” pair (different from Darlington pair)
- ❖ $A_v \approx 1$

Class AB output of MOSFET op-amp



- ❖ Improved output stage cannot drive low impedance load, say 8Ω speaker.

Output Short-Circuit Current (Source: MCP6001 datasheet)	I_{SC}	—	± 6	—	mA	$V_{DD} = 1.8V$
		—	± 23	—	mA	$V_{DD} = 5.5V$

Op-amp Output Specifications

- ❖ These are the output specifications for the MCP6001 op-amp.

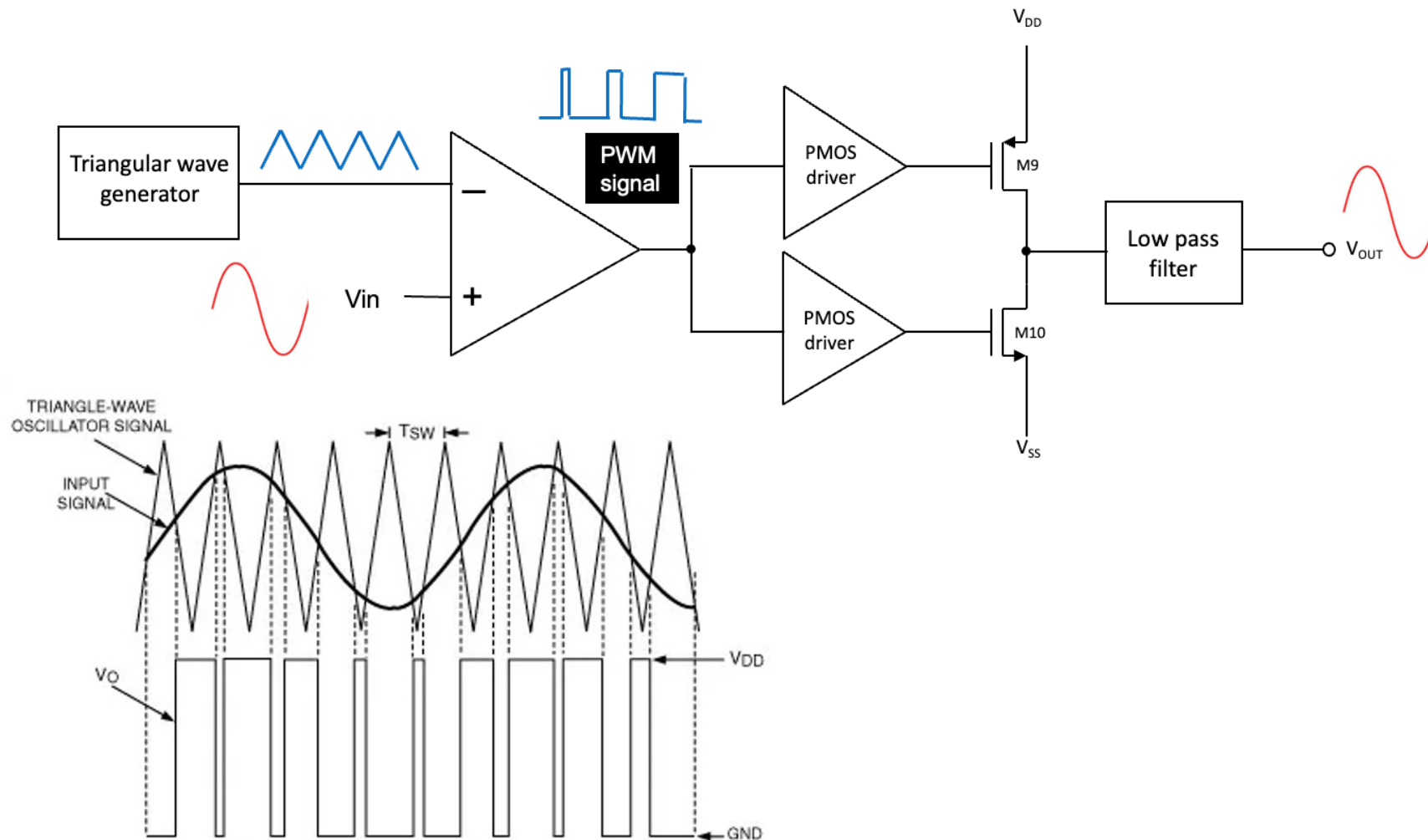
Output						
Maximum Output Voltage Swing	V_{OL}, V_{OH}	$V_{SS} + 25$	—	$V_{DD} - 25$	mV	$V_{DD} = 5.5V$, 0.5V input overdrive
Output Short-Circuit Current	I_{SC}	—	± 6	—	mA	$V_{DD} = 1.8V$
		—	± 23	—	mA	$V_{DD} = 5.5V$



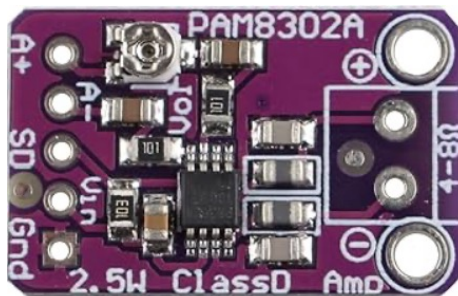
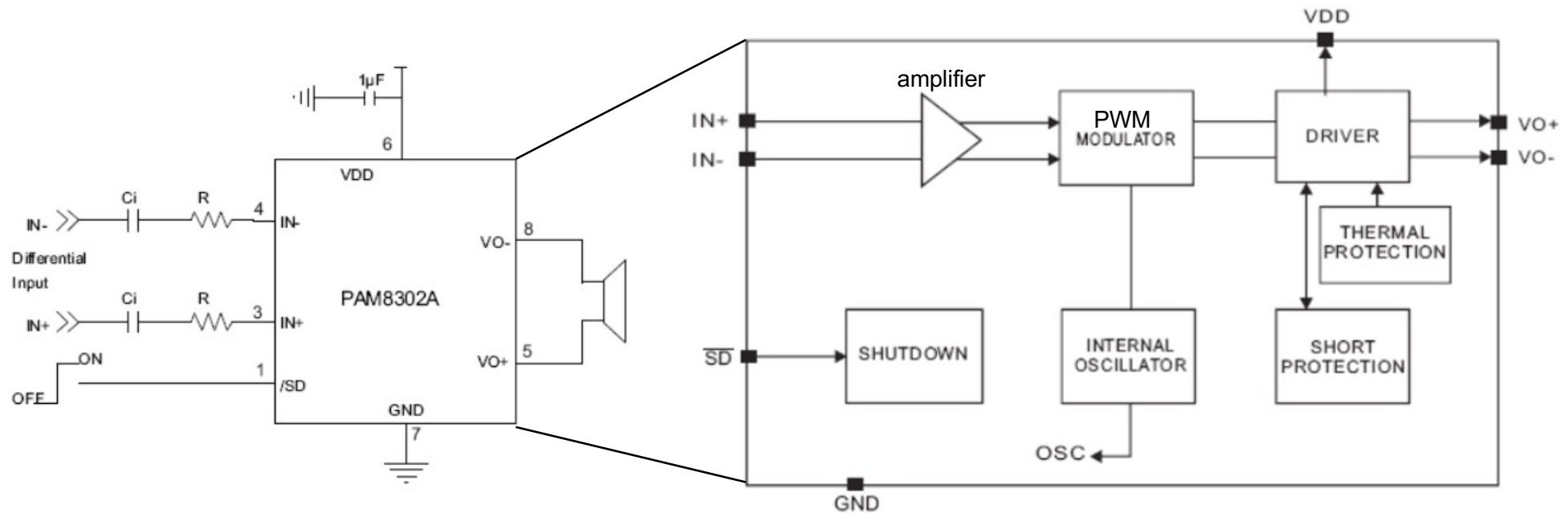
Highlighted entries are implemented in the MCP6001 LTSpice model.

Source: MCP6001 [datasheet](#)

Block Diagram of a Class D amplifier



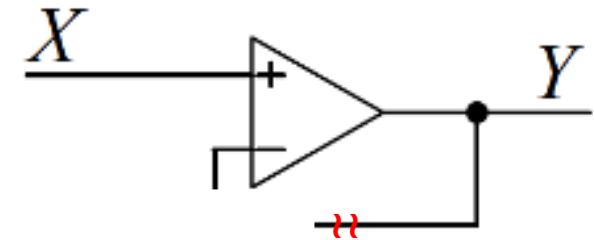
The PAM8302A Audio Amplifier



- 2.5W Output at 10% THD with a 4Ω Load and 5V Power Supply
- Filterless, Low Quiescent Current and Low EMI
- High Efficiency up to 88%
- Superior Low Noise
- Short Circuit Protection

Stability issue in op-amps

- ❖ Op-amp as an amplifier always uses **NEGATIVE FEEDBACK** to determine the gain.
- ❖ Consider this unity gain buffer: break the loop in the feedback path.
- ❖ Due to delay, could feedback a signal that is 180° out of phase relative to input.
- ❖ The negative feedback now become positive feedback, leading to oscillation.

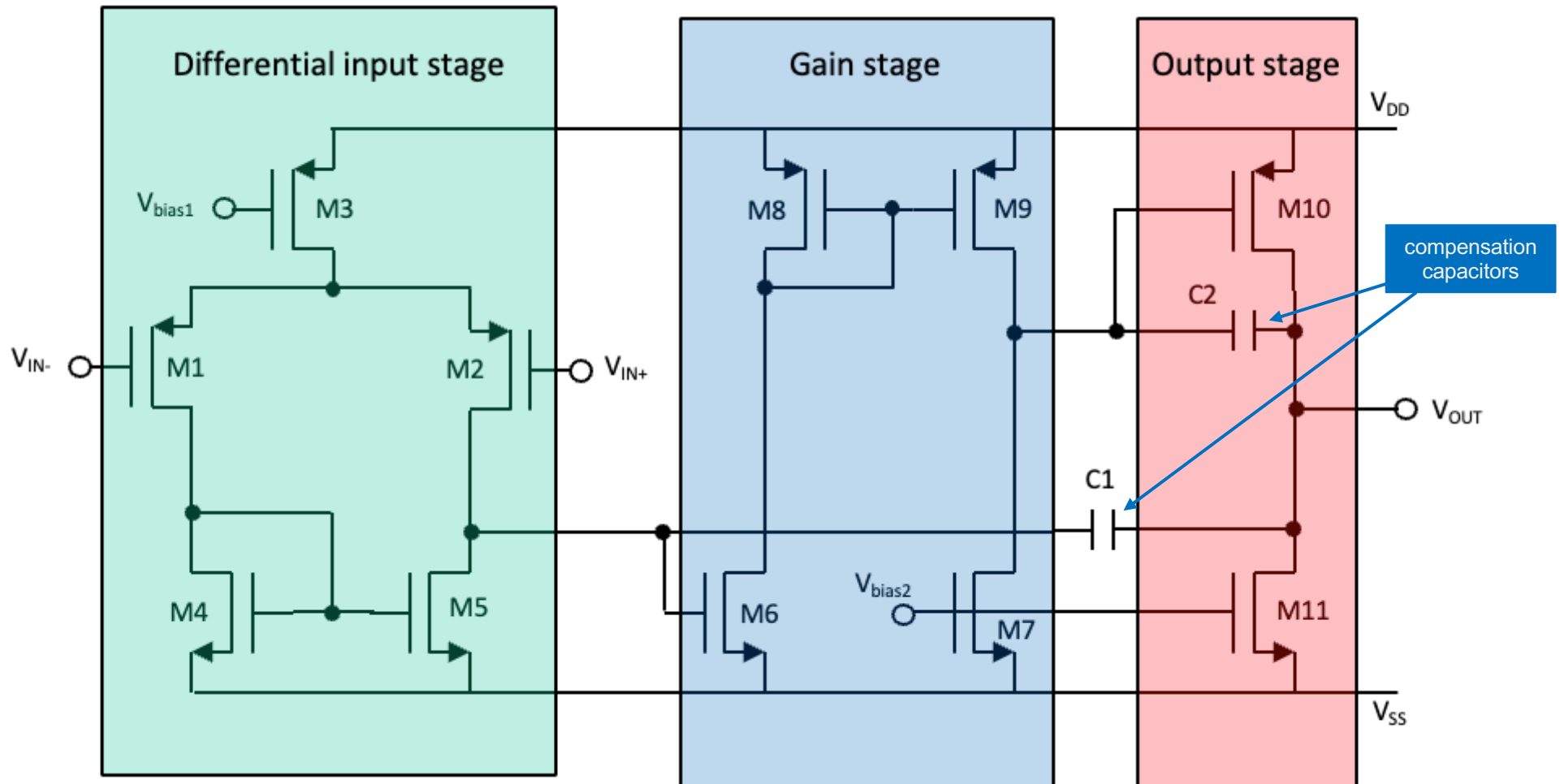


Gain = 1

Stability requirement:

1. The open-loop **gain is > 1** , but the **phase angle** must be **$< 180^\circ$** at **all signal frequencies**, OR
2. The **phase angle** is **$\geq 180^\circ$** , but the open-loop **gain is < 1** at **all signal frequencies**.

Open-loop gain and compensation capacitors



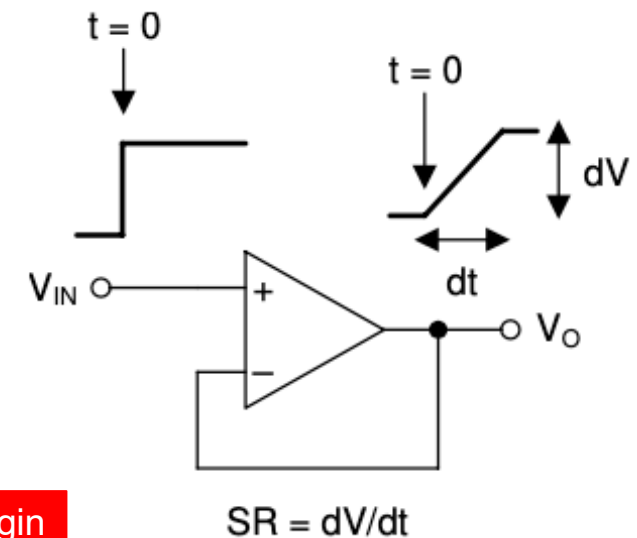
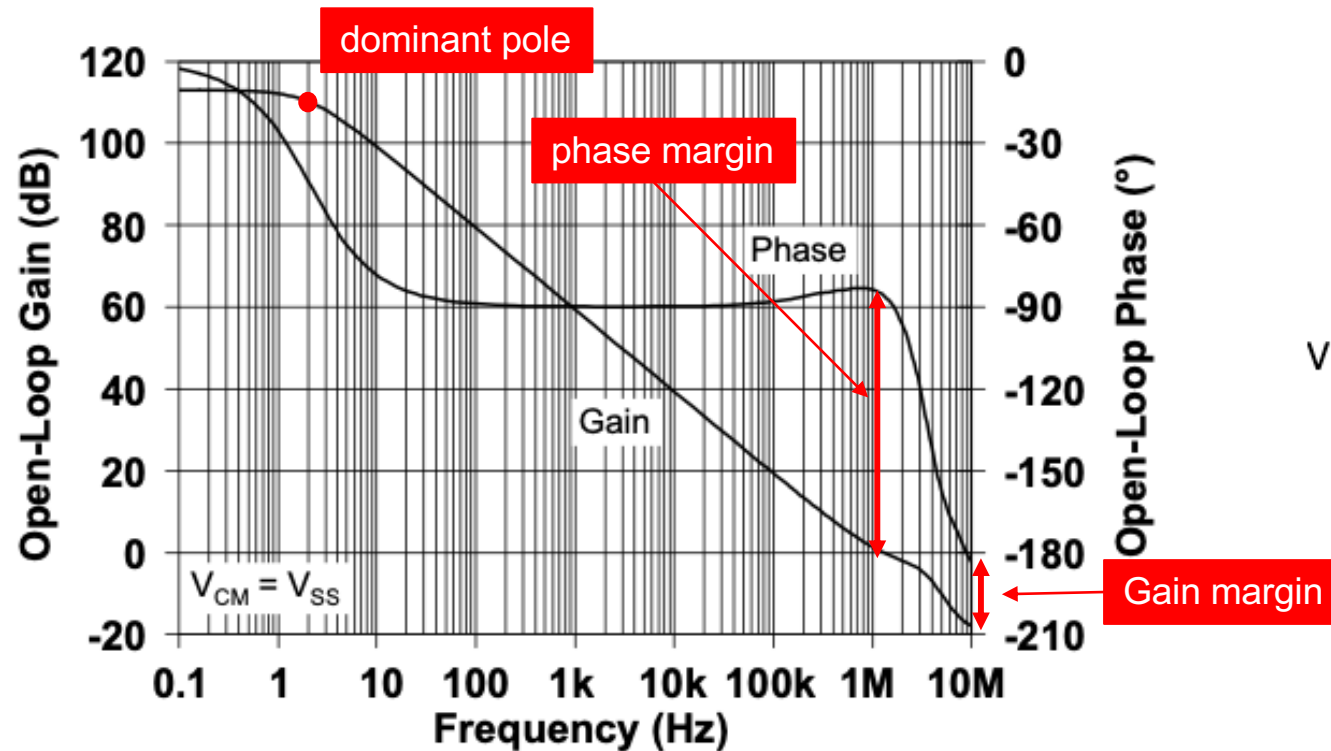
Op-amp Loop Gain specification

Open-Loop Gain

DC Open-Loop Gain (Large Signal)	A_{OL}	88	112	—	dB	$V_{OUT} = 0.3V \text{ to } V_{DD} - 0.3V,$ $V_{CM} = V_{SS}$
----------------------------------	----------	----	-----	---	----	--

AC Response

Gain Bandwidth Product	GBWP	—	1.0	—	MHz	
Phase Margin	PM	—	90	—	°	$G = +1 \text{ V/V}$
Slew Rate	SR	—	0.6	—	V/ μ s	



Source: MCP6001 [datasheet](#)